

Original

Minimum Enclosing Circle

Consider the following problem:

Library Checker - Minimum Enclosing Circle

You're given $n \leq 10^5$ points $p_i = (x_i, y_i)$.

For each p_i , find whether it lies on the circumference of the minimum enclosing circle of $\{p_1, \dots, p_n\}$.

Here, by the minimum enclosing circle (MEC) we mean a circle with minimum possible radius that contains all the n points, inside the circle or on its boundary. This problem has a simple randomized solution that, on first glance, looks like it would run in $O(n^3)$, but actually works in $O(n)$ expected time.

To better understand the reasoning below, we should immediately note that the solution to the problem is unique:

Why is the MEC unique?

Consider the following setup: Let r be the radius of the MEC. We draw a circle of radius r around each of the points $p_1, \dots, p_n$. Geometrically, the centers of circles that have radius r and cover all the points $p_1, \dots, p_n$ form the intersection of all n circles.

Now, if the intersection is just a single point, this already proves that it is unique. Otherwise, the intersection is a shape of non-zero area, so we can reduce r by a tiny bit, and still have non-empty intersection, which contradicts the assumption that r was the minimum possible radius of the enclosing circle.

With a similar logic, we can also show the uniqueness of the MEC if we additionally demand that it passes through a given specific point p_i or two points p_i and p_j (it is also unique because its radius uniquely defines it).

Alternatively, we can also assume that there are two MECs, and then notice that their intersection (which contains the points $p_1, \dots, p_n$ already) must have a smaller diameter than initial circles, and thus can be covered with a smaller circle.